

Osmania University

MID-TERM EXAMINATION ? 2025

Basic Electrical Engineering ? Mid-Term Examination, 2025

Subject Code: EE202 Semester: 2 Regulation: R18

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: DC Circuits)

1. Describe the working principle of DC Circuits with necessary equations. [12 Marks]
2. Solve a problem related to DC Circuits. Show all the steps clearly. [12 Marks]
3. Solve a problem related to DC Circuits. Show all the steps clearly. [12 Marks]

SECTION B (Unit 2: AC Circuits)

4. List the advantages and limitations of AC Circuits. Explain any two in detail. [12 Marks]
5. Solve a problem related to AC Circuits. Show all the steps clearly. [8 Marks]
6. How is AC Circuits implemented? Give an algorithm / procedure and explain. [12 Marks]

SECTION C (Unit 3: Transformers)

7. Describe the working principle of Transformers with necessary equations. [12 Marks]
8. Derive the expression related to Transformers and explain each term. [8 Marks]
9. Discuss the engineering applications of Transformers. [8 Marks]

SECTION D (Unit 4: Electrical Machines)

10. Solve a problem related to Electrical Machines. Show all the steps clearly. [12 Marks]
11. Describe the working principle of Electrical Machines with necessary equations. [12 Marks]
12. Define Electrical Machines. Explain its significance in Basic Electrical Engineering with a suitable example. [8 Marks]

SECTION E (Unit 5: Measuring Instruments)

13. Explain Measuring Instruments with a neat diagram/sketch and a suitable example. [12 Marks]
14. Derive the expression related to Measuring Instruments and explain each term. [12 Marks]
15. Describe the working principle of Measuring Instruments with necessary equations. [12 Marks]

***** End of Question Paper *****